

WASP-36b

R-0813

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_240112 · created 2026-07-28T12:18:37+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460321.8325 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1623 +/- 0.0082
Transit depth (Rp/Rs)^2	2.63 +/- 0.27 [%]
Orbital Inclination (inc)	83.49 +/- 0.74
Airmass coefficient 1 (a1)	1.0503 +/- 0.0099
Airmass coefficient 2 (a2)	-0.0353 +/- 0.01
Scatter in the residuals of the lightcurve fit is	0.94 %
Best Comparison Star	#1 - [349, 213]
Optimal Aperture	3.7
Optimal Annulus	13.85
Transit Duration (day)	0.0804 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1623 ± 0.0082 vs 0.1368 (deviation 2.84σ)
Duration fit vs published	0.0804 d vs 0.0758 d (deviation 0.93σ)
Mid-time O–C	-3.5 min
Depth SNR	18.1
Residual scatter	0.94 %

Light curve

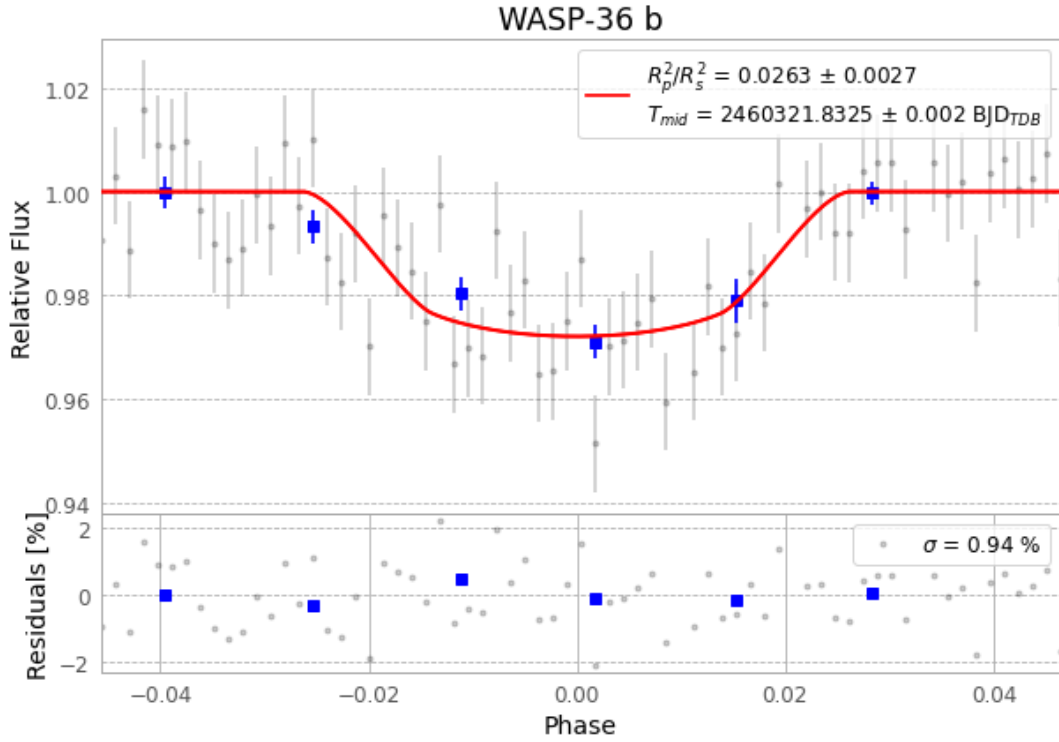


Figure 1 — FinalLightCurve_WASP-36 b_2024-01-11.png (SHA-256 1ed359d2e3828c51...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [435, 260], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2e006bdee0d9cb94...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

69 FITS frames (45 MB), WASP-36240112060915.FITS → WASP-36240112093315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-240112.log (7452c56b6f41...), FinalLightCurve_WASP-36 b_2024-01-11.png (1ed359d2e382...), FinalLightCurve_WASP-36 b_2024-01-11.csv (9cc5157de9b2...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 12:18Z.